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Arnaud Blouin & Benoit Combemale (Univ Rennes, IRISA, Inria)

Philippe Palanque (IRIT)

DiverSE research group

Subject: Application for PhD Position – A software engineering approach for interactive digital twins (PC5)

Dear Arnaud Blouin, Benoit Combemale, and Philippe Palanque,

I am writing to apply for the PC5 PhD position regarding a software engineering approach for interactive digital twins, hosted at the DiverSE research group. With a strong background in software engineering, component-based architectures, and human-computer interaction (HCI), I am highly motivated to address the challenge of decoupling interactive UI components from complex digital twin simulations.

A central problem in current digital twin development is the rigid coupling between simulation models and user interfaces, which limits reusability and scalability. During the development of the "Live Control Room," I architected a modern Next.js and Three.js 3D web dashboard designed explicitly around separation of concerns. By engineering a decoupled component hierarchy, I ensured that the interactive visualizations (which managed user commands and state exploration) communicated with the underlying physical simulations exclusively through robust, standardized APIs. This hands-on experience in modern frontend software engineering positions me perfectly to develop the model-driven engineering (MDE) frameworks you require. I am eager to research novel domain-specific languages (DSLs) that can abstract UI interactions and automate the generation of interactive digital twin interfaces across various hardware configurations.

I am particularly excited about validating these software engineering approaches on the industrial use cases within the Artemis framework. Please find attached my resume and academic transcripts, which detail my academic background and practical project implementations. I welcome the opportunity to discuss my application further.

Sincerely,

Ismail Aissaoui